

Data Collection and Exploratory Data Analysis

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Outline

- ① Data Sources and Types
- ② Why Exploratory Data Analysis?
- ③ One Variable at a Time

Data Sources and Types

Learning outcomes

- Explain why exploratory data analysis (EDA) comes before modelling
- Choose summaries and plots that suit the variable type and the question
- Read distributions, relationships, and structure in real data, and interpret them in context
- Recognise when EDA already answers the question, and when it shows that a model is needed

Roadmap: data and variable types → why EDA → one variable → data quality → two variables → many variables → time and space → when EDA is enough → pitfalls

Where data come from

Source	Examples	What EDA must check
Surveys, administrative records	DHS, MICS, census extracts, hospital or school records	design weights, codes for “don’t know”, top-coding
Sensors and logs	weather stations, phones, server logs	timestamps, gaps, drift
Web and APIs	scraped pages, social media, open data	duplicates, selection bias
Remote sensing	satellite imagery, NDVI	resolution, cloud cover, alignment

- Microdata means one row per person, household, or event rather than pre-aggregated tables
- The source determines who is included, how variables were measured, and what can go wrong

How data are structured

- **Structured:** fixed rows and columns with a schema, such as survey microdata, spreadsheets, and relational tables
- **Semi-structured:** self-describing hierarchy without a fixed table layout, such as JSON from APIs, XML, and server logs; it must be parsed into columns before analysis
- **Unstructured:** no predefined layout, such as free text, images, audio, and video; useful numbers must be extracted first
- This lecture works with structured (tabular) data; the other types enter once their content has been turned into columns

Variable types decide the analysis

Type	Example	Summaries	Plots
Nominal	sex, embarkation port	counts, mode	bar chart
Ordinal	diamond cut, class	counts, median	ordered bars
Discrete	number of children	counts, mean	bar chart
Continuous	price, age	mean, median, SD	histogram, boxplot

- The type, not the storage, decides which summary is meaningful: class is stored as 1, 2, 3, yet an average class of 2.3 means nothing
- Survey microdata often store categories as numeric codes (1 = male, 2 = female, 99 = don't know); recognise them before summarising

Why Exploratory Data Analysis?

What is EDA?

- Exploratory data analysis (EDA) is the systematic examination of a dataset through summary statistics and graphics before any formal modelling, in the spirit of John Tukey's "detective work".
- It asks what each variable looks like, how variables relate to each other, which values are unusual, and whether there is structure such as groups or trends.
- Its output is decisions: which questions the data can answer, which problems must be fixed first, and which kinds of model, if any, are worth trying.
- EDA is not confirmatory: patterns found this way are hypotheses, to be tested on data that were not used to find them.

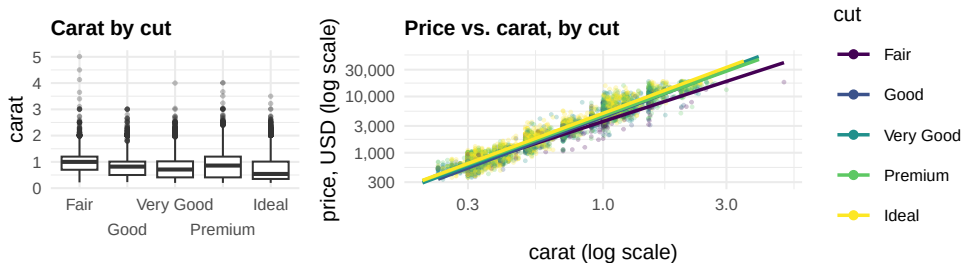
Why look at the data? An example

Diamonds: 53,940 stones, price in US dollars. Mean price by cut, from worst to best:

Cut	n	Mean price	Mean carat
Fair	1,610	\$4,359	1.05
Good	4,906	\$3,929	0.85
Very Good	12,082	\$3,982	0.81
Premium	13,791	\$4,584	0.89
Ideal	21,551	\$3,458	0.70

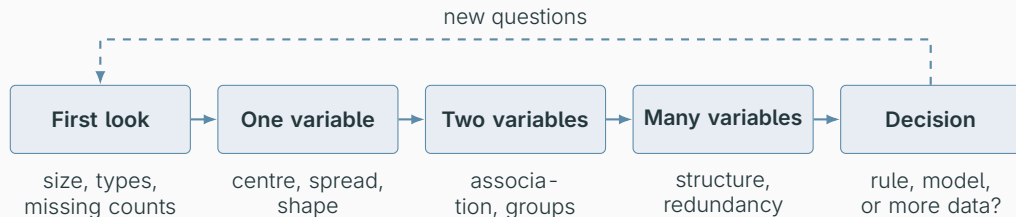
- The best cut (Ideal) has the lowest mean price, and the worst cut (Fair) is dearer than Good, Very Good, and Ideal
- Taken at face value, a better cut lowers the price, and a model using cut alone would learn exactly that
- The last column hints at the reason: Ideal stones are the smallest on average (0.70 carat) and Fair the largest (1.05)

Why look at the data? The plots



- Left: Ideal stones are mostly small (median 0.54 carat) and Fair stones large (median 1.00 carat)
- Right: within every cut, price rises steeply with carat; at exactly 1 carat, Fair costs \$4,130 on average ($n = 166$) and Ideal \$5,547 ($n = 208$)
- Cut and carat are entangled: the table by cut mixed two effects, and looking at the variables jointly separated them

The EDA workflow



- Each stage raises questions for the next, so EDA is iterative rather than a fixed pipeline
- The decision at the end is not always “fit a model”: it may be to collect more data, apply a simple rule, or report the summary

First look at a dataset

- Size and types: number of rows and columns and the type of each variable (`glimpse(df)` in R, `df.info()` in pandas)
- Missing values: count them per variable (`colSums(is.na(df))` , `df.isna().sum()`); if any are present, judge whether action is required before analysis, and take the necessary steps (treatments are covered in the data preprocessing lecture)
- Duplicate rows and impossible values, such as negative ages or zero dimensions
- Explore the training data only; looking at the test set leaks information into modelling decisions
- Titanic (891 passengers): age is missing for 177 passengers, deck for 688, and port of embarkation for 2

One Variable at a Time

Location: mean, median, trimmed mean, mode

Observations $x_1, \dots, x_n$ of one numeric variable.

- **Mean** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$: uses every value, so extreme values pull it
- **Median**: the middle value of the sorted data, with half of the observations on each side; extreme values do not move it, so it is the better measure of centre for skewed data such as prices, incomes, or waiting times
- **Trimmed mean**: the mean after discarding a fixed share (say 10%) of observations from each tail; a compromise between the two
- **Mode**: the most frequent value; the natural centre of a categorical variable, and for a continuous variable it is read from the peak of a histogram or density

Spread and shape

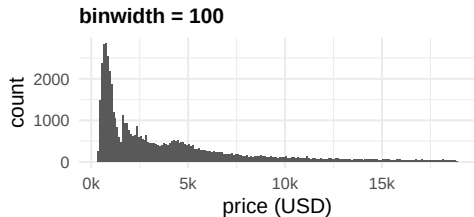
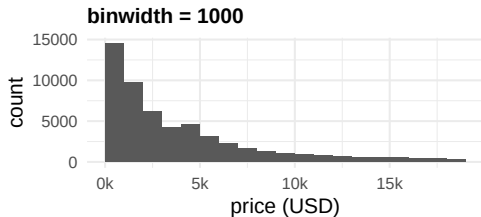
- **Standard deviation** $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$: typical distance from the mean; sensitive to extreme values
- **Interquartile range** $IQR = Q_3 - Q_1$: width of the middle 50% of the data; robust to extreme values
- **Median absolute deviation** $MAD = \text{median}_i |x_i - \tilde{x}|$, where $\tilde{x}$ is the median (no rescaling): a robust alternative to the standard deviation
- **Skewness** $g_1 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s}\right)^3$: positive for a long right tail, negative for a long left tail, near zero for a symmetric distribution
- **Excess kurtosis** $g_2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$: positive when the tails are heavier than those of a normal distribution

Summary statistics in practice: diamonds

	price (USD)	carat
Mean	3,933	0.80
10% trimmed	3,159	0.73
Median	2,401	0.70
Std. dev.	3,989	0.47
IQR	4,374	0.64
MAD	1,670	0.32
Skewness	1.62	1.12
Ex. kurtosis	2.18	1.26

- The mean price is 64% above the median: a few expensive stones pull the mean up, while the median describes a typical stone
- Skewness of 1.62 confirms a long right tail; mean > median and positive skewness go together
- Mean – SD is –\$57, an impossible price: mean $\pm$ SD is a poor summary here, and quartiles are better

Histograms and density estimates



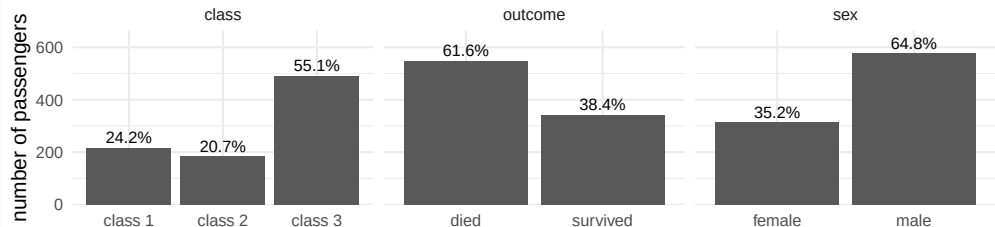
- **Histogram:** counts in equal-width bins; try several widths (wide bins hide structure, narrow bins add noise); read peaks, gaps, and spikes
 - At width 100 the count falls to 463 in \$1,500–1,599 and jumps to 1,115 in the next bin, a jump that width 1000 hides
 - Use with hundreds of observations or more; for small samples use a dot or strip plot

Boxplot, violin plot, and ECDF



- **Boxplot:** box from Q_1 to Q_3 with the median line; whiskers reach the furthest points within $1.5 \times \text{IQR}$; points beyond are flagged (3,540 stones, 6.6%, all expensive)
 - Use to compare many groups side by side; it cannot show several peaks

Categorical variables: frequencies and bar charts



- A frequency table lists the count and share of each level, and the bar chart is its picture
- Titanic, 891 passengers: 61.6% died and 38.4% survived; 55.1% travelled third class; 64.8% were male
- Imbalance matters: predicting “died” for everyone is already 61.6% accurate, a baseline any model must beat
- Order nominal levels by frequency and ordinal ones naturally; avoid pie charts

Questions?
